

Question ID: 70753f99

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function f is defined by $f(x) = (x + 3)(x + 1)$. The graph of f in the xy -plane is a parabola. Which of the following intervals contains the x -coordinate of the vertex of the graph of f ?

Answer

- A. $-4 < x < -3$
- B. $-3 < x < 1$
- C. $1 < x < 3$
- D. $3 < x < 4$

Correct Answer: B

Rationale

Choice B is correct. The graph of a quadratic function in the xy -plane is a parabola. The axis of symmetry of the parabola passes through the vertex of the parabola. Therefore, the vertex of the parabola and the midpoint of the segment between the two x -intercepts of the graph have the same x -coordinate. Since $f(-3) = f(-1) = 0$, the x -coordinate of the vertex is $\frac{(-3) + (-1)}{2} = -2$. Of the shown intervals, only the interval in choice B contains -2 . Choices A, C, and D are incorrect and may result from either calculation errors or misidentification of the graph's x -intercepts.